

Restricted Three-Body Problem

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Abstract—This is an assignment report of the "Celestial Mechanics" course given by Asst. Prof. David M. Hernandez at National Taiwan Normal University. This report aims to investigate the complex orbital dynamics within the Circular Restricted Three-Body Problem (CR3BP). The study utilizes Leapfrog numerical integration techniques to solve the equations of motion for a particle of negligible mass moving under the gravitational influence of two larger primary bodies. By calculating the Jacobi constant and evaluating specific energy thresholds, we classify distinct orbital regimes, ranging from periodic resonant and tadpole orbits to unstable chaotic and escape trajectories. The analysis specifically focuses on mapping the stability of Lagrange points and demonstrating how initial velocity conditions dictate the final path of the particle. These findings provide a practical understanding of how gravitational mechanics can be applied to space mission design and orbital maintenance, illustrating the clear distinction between stable, trapped orbits and escape paths.

I. INTRODUCTION

IN celestial mechanics, understanding how objects move through space is a fundamental task. Historically, the two-body problem, which describes the motion of two masses moving under their mutual gravitational attraction, provided clear, analytical results. We know that in this system, the motion is predictable, and the paths of the bodies are closed geometrical shapes within inertial space. However, adding a third body makes the situation much more complicated. This complexity is the central theme of the Circular Restricted Three-Body Problem (CR3BP).

A. Context

The CR3BP is a specific, simplified version of the general three-body problem. In this model, we consider three bodies: two primary objects and a third object that has negligible mass compared to the other two. Because the third body is so small, it does not influence the motion of the two massive primaries. We assume that the two massive bodies move in circular, coplanar orbits around their common centre of mass. By making these assumptions, we create a system where we can analyze the gravitational interaction without the extreme mathematical difficulty found in the general version of the problem.

The mathematical challenge of the three-body problem has fascinated scientists for hundreds of years. Because the problem is so difficult to solve completely, it has become a famous topic for many important figures in mathematics and physics. Notable scholars, including Euler, Lagrange, Laplace, Jacobi, Le Verrier, Hamilton, Poincaré, and Birkhoff, have all made major contributions to our understanding of

this subject. Their work laid the necessary foundation for studying orbital dynamics. While authoritative books by authors such as Szebehely (1967) and Marchal (1990) provide deep insights into the literature and derivations of important results, the problem remains enigmatic.

While the problem is classic, it is not just a historical curiosity. Recent developments in nonlinear dynamics have brought new attention to the field. Modern observations of the solar system have shown us complex orbital structures that require better models. Consequently, there is a renewed interest in exploring the gravitational interactions within the CR3BP. By studying these systems, we gain insights into how particles move in complex gravitational fields, which is essential for understanding satellite trajectories, planetary ring systems, and the long-term stability of the solar system. This report investigates these dynamics by applying the mathematical framework derived from classical celestial mechanics, focusing on how initial conditions and mass ratios define the behavior of the third body.

B. Objectives

The primary objective of this project is to investigate the complex dynamical behavior of the Circular Restricted Three-Body Problem (CR3BP) through systematic numerical analysis. Central to this study is the integration of the non-linear equations of motion, as derived in Murray & Dermott, to simulate the trajectories of a massless test particle under the gravitational influence of two primary bodies.

Building upon this numerical foundation, the project aims to achieve the following specific goals:

- 1 **Orbit Classification:** We seek to identify, visualize, and categorize a diverse range of orbital classes. By varying the initial conditions $(x_0, y_0, \dot{x}_0, \dot{y}_0)$, we intend to map the resulting structures within the phase space, distinguishing between bounded, transit, horseshoe, tadpole, and chaotic trajectories.
- 2 **Energy Regime Analysis:** A major goal is to investigate how the Jacobi constant (C_J) influences the motion of the test particle. We will explore the orbital behavior across different regimes ($C_J < U_1$ through $C_J \geq U_4 = U_5$), determining the "allowed" regions of motion for each case and interpreting the physical constraints imposed by these zero-velocity surfaces.
- 3 **Lagrange Point Stability:** Finally, we will rigorously assess the stability of the equilibrium points. Specifically, we aim to verify the analytical stability criterion for

the Lagrange points as defined by the mass ratio (μ) threshold. We will compare these theoretical predictions with our numerical results to discuss the transition between stable and unstable behavior.

Through this methodology, this project intends to provide a comprehensive characterization of the CR3BP, illustrating the delicate interplay between mass distribution, energy states, and the resulting orbital topology.

C. Physical Parameters

To generalize the analysis of the circular restricted three-body problem for any pair of massive bodies, we utilize a fundamental non-dimensional parameter known as the mass ratio, denoted as μ . This parameter allows us to normalize the gravitational system, effectively reducing the equations of motion to a dimensionless form where the distance between the two primary bodies is set to unity.

The mass ratio is defined as:

$$\mu = \frac{m_1}{m_0 + m_1}$$

In this expression, m_0 and m_1 represent the masses of the two primary bodies in the system. By employing this mass ratio, we can compare different gravitational configurations—such as the Earth-Moon system or the Sun-Jupiter system—without needing to scale the physical units for every calculation.

The parameter μ is the primary determinant of the gravitational potential landscape. Variations in this value fundamentally alter the positions of the Lagrange equilibrium points and determine the specific energy thresholds (Jacobi constant regimes) that dictate whether a test particle is trapped within a specific orbital region or permitted to move between them. Throughout this report, we will systematically explore how different values of μ influence the stability of orbits and the overall structure of the phase space available to the third body.

II. METHODOLOGY

A. Equations

To analyze the motion of a particle with negligible mass within the circular restricted three-body problem, we use a rotating coordinate system, known as the synodic frame. This frame shares the same origin as the inertial system but rotates at a constant angular velocity n . By choosing this rotating frame, the positions of the two primary masses (m_1 and m_2) become fixed, which significantly simplifies the study of the test particle's trajectory.

In this system, the motion of the particle is described by three second-order differential equations. These equations incorporate the gravitational influence of the primary bodies, as well as the fictitious forces, specifically the Coriolis and centrifugal accelerations, that arise because our frame of reference is rotating.

The governing equations of motion in the synodic frame are defined as:

$$\begin{aligned}\ddot{x} - 2n\dot{y} - n^2x &= - \left[\mu_1 \frac{x + \mu_2}{r_1^3} + \mu_2 \frac{x - \mu_1}{r_2^3} \right] \\ \ddot{y} + 2n\dot{x} - n^2y &= - \left[\frac{\mu_1}{r_1^3} + \frac{\mu_2}{r_2^3} \right] y \\ \ddot{z} &= - \left[\frac{\mu_1}{r_1^3} + \frac{\mu_2}{r_2^3} \right] z\end{aligned}$$

In these equations, $\ddot{x}$, $\ddot{y}$, and $\ddot{z}$ represent the accelerations of the particle in the rotating frame. The terms $2n\dot{y}$ and $2n\dot{x}$ denote the Coriolis accelerations, while the terms involving n^2x and n^2y represent the centrifugal accelerations.

The variables r_1 and r_2 represent the distances from the test particle at position (x, y, z) to the primary mass (μ_1) and secondary mass (μ_2), respectively. These distances are defined by the following relations:

$$\begin{aligned}r_1^2 &= (x + \mu_2)^2 + y^2 + z^2 \\ r_2^2 &= (x - \mu_1)^2 + y^2 + z^2\end{aligned}$$

These equations allow us to calculate the path of a test particle for any given set of initial conditions, providing the necessary mathematical foundation for classifying the orbital regimes discussed in the subsequent sections of this report.

B. Lagrange Points and Stability

In the circular restricted three-body problem, we identify special positions where a test particle can remain stationary relative to the two primary masses, m_1 and m_2 , within the rotating synodic frame. At these points, the particle has zero velocity and zero acceleration in the rotating system, meaning the net gravitational and centrifugal forces are perfectly balanced. These specific locations are known as Lagrangian equilibrium points.

Mathematically, these points are found by setting the partial derivatives of the scalar function U (the pseudo-potential) with respect to x and y to zero. The specific equations for finding these locations are:

$$\frac{\partial U}{\partial x} = 0, \quad \frac{\partial U}{\partial y} = 0$$

These equations yield five distinct solutions:

- 1 Collinear Points (L_1, L_2, L_3): Three of these equilibrium points lie along the x-axis, which is the line connecting the two primary masses. The L_1 point is situated between the two masses, while L_2 is found outside the secondary mass, and L_3 lies on the negative x-axis.
- 2 Triangular Points (L_4, L_5): The remaining two points are the triangular Lagrangian equilibrium points, which form the apexes of equilateral triangles with the two primary masses. As derived in the text, these points are located at:

$$x = \frac{1}{2} - \mu_2, \quad y = \pm \frac{\sqrt{3}}{2}$$

Regarding the stability of these points, the triangular Lagrange points (L_4 and L_5) are of particular interest as they can be stable under specific mass ratios. The stability of these points depends on the mass ratio μ of the system. According to the criteria provided in the project brief, the triangular Lagrange points are stable if the mass ratio satisfies the following inequality:

$$\mu < \frac{1 - \sqrt{23/27}}{2}$$

If the mass ratio exceeds this value, the points become unstable. This threshold is a fundamental physical constraint that determines whether a small particle can maintain a stable orbit near these triangular equilibrium positions or if it will be perturbed away by the gravitational influence of the primaries.

C. Numerical Integration: Leapfrog

To find the path of the test particle, we solve the equations of motion numerically. The project guidelines permit the use of any numerical integrator, provided that the integration parameters are suitable and the error in the Jacobi constant remains low. For this report, we have implemented the Leapfrog integrator to calculate the trajectories of the test particle.

The Leapfrog method is a type of symplectic integrator. Symplectic integrators are particularly useful in celestial mechanics because they preserve the Hamiltonian structure of the system. In practice, this means that the method is highly effective at keeping the Jacobi constant stable over long periods of time, preventing the artificial energy drift that can occur with simpler numerical methods.

To begin the simulation, we must define the state of the particle at the starting time ($t = 0$). As specified in the project brief, the system requires four initial-condition parameters, including initial position (x_0, y_0) and initial velocity ($\dot{x}_0, \dot{y}_0$).

The Leapfrog algorithm computes the trajectory by updating the particle's position and velocity in a repeating, alternating pattern. Specifically, the method updates the position based on the current velocity, and then updates the velocity based on the forces acting at the new position. This process continues in small time steps (Δt).

By using this leapfrog pattern, we ensure that the position and velocity updates are staggered in time. This characteristic helps to minimize numerical errors in the trajectory.

D. Jacobi Constant Analysis

In the circular restricted three-body problem, the total energy of the test particle is not conserved in the traditional sense because the coordinate system is rotating. However, the system possesses a specific integral of motion known as the Jacobi Constant (C_J). This constant is essential for understanding the dynamics of the system, as it determines the regions in space where the particle is physically allowed

to exist.

The test particle moves within an effective potential field, U . This potential combines the gravitational influence of the two primary masses with the centrifugal force caused by the rotation of the synodic coordinate frame. The Jacobi Constant relates the particle's energy to this potential and its velocity, v , through the following relationship:

$$C_J = 2U - v^2$$

Because the square of the particle's velocity (v^2) must always be a positive value or zero, the particle is restricted to regions where the potential is greater than or equal to the Jacobi Constant ($2U \geq C_J$). The boundary where $2U = C_J$ is referred to as the "zero-velocity curve." If a particle's trajectory reaches this curve, its velocity becomes zero, and it must reverse its direction. Therefore, these curves function as barriers, or "gates," that can trap the particle within specific areas of the system or block it from reaching certain regions.

The accessibility of these regions is determined by the value of the Jacobi Constant relative to the potential values at the five Lagrange points. As in this project, we must investigate the orbital behavior across five specific regimes: $C_J < U_1$, $U_1 < C_J < U_2$, $U_2 < C_J < U_3$, $U_3 < C_J < U_4 = U_5$, $C_J \geq U_4 = U_5$.

To determine the transitions between these regimes, we calculate the values of the Jacobi Constant at the Lagrange points, denoted as C_{Li} . According to Murray & Dermott (Eqs. 3.94–3.98), the Jacobi Constant values at these critical equilibrium points can be approximated using the mass ratio, μ :

$$C_{L1} \approx 3 + 3^{4/3}\mu^{2/3} - 10\mu/3$$

$$C_{L2} \approx 3 + 3^{4/3}\mu^{2/3} - 14\mu/3$$

$$C_{L3} \approx 3 + \mu^2$$

$$C_{L4} \approx 3 - \mu^2$$

$$C_{L5} \approx 3 - \mu^2$$

When the Jacobi Constant of a particle is higher than the value at a specific Lagrange point, the zero-velocity surfaces are closed, meaning the particle is trapped within a confined region and cannot pass through that Lagrange point. Conversely, as the Jacobi Constant decreases (implying the particle has higher energy), these surfaces open at the Lagrange points, allowing the particle to transit into new regions of space. By comparing the calculated C_J of our trajectories against these critical C_{Li} thresholds, we can determine whether a particle is restricted to orbits around a single primary or if it is free to travel throughout the entire system.

III. RESULTS AND DISCUSSION

A. Regime Mapping

To understand how a test particle moves in the circular restricted three-body problem, we mapped the allowed regions of motion using the Jacobi Constant (C_J). As discussed in the methodology, the Jacobi Constant acts as a barrier because of the relationship $2U - v^2 = C_J$. This means that the particle can only exist in regions where $2U \geq C_J$. By varying C_J and comparing it to the potential values (U_i) at the Lagrange points, we identified specific regimes that dictate whether the particle is free to roam the entire system or trapped within a small area.

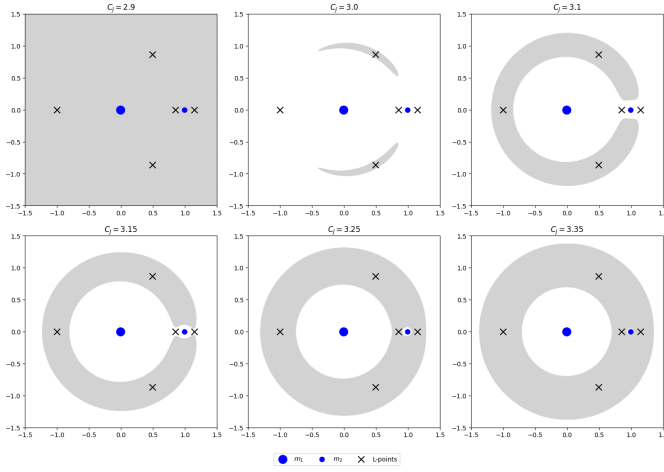


Fig. 1: This diagram illustrates the zero-velocity curves for a test particle at various energy levels (Jacobi constants). The shaded regions represent areas where the particle is forbidden to travel. As the Jacobi constant (C_J) increases, the gateways at the Lagrange points (L_1 – L_5) gradually close, trapping the particle in specific regions.

The results of the mapping confirm that the Jacobi Constant is the most important factor in determining the behavior of the test particle. As shown in Figure 1, the zero-velocity curves act as physical boundaries. These boundaries effectively divide the coordinate plane into distinct chambers.

At low energy level, the Jacobi constant is lower than the potential at the L_1 point. Because C_J is very low, the gateways at the Lagrange points are wide open. In this regime, the test particle is not restricted. It can move freely between the primary and secondary masses, as there are no physical barriers to stop its motion. As we increase the Jacobi constant, the energy barriers begin to rise, and eventually, the gate at L_1 closes. The particle can no longer pass through the area between the two masses easily. As C_J continues to increase, the gates at L_2 and L_3 also close. This separates the space into different, isolated chambers. And continue like that, the particle becomes increasingly confined. It might be trapped in a region surrounding one of the primary masses. It cannot pass through the Lagrange points to reach the other mass, meaning its orbit is restricted to the vicinity of the mass it was

originally near. At high energy level, the barriers are at their most restrictive. The only remaining open areas are often the small regions around the triangular Lagrange points L_4 and L_5 . If the particle is in this regime, it is essentially locked into a specific area, and its motion is strictly limited by the surrounding potential walls.

B. Initial Parameters Sensitivity

1) *Initial Positions:* In this section, we analyze how the behavior of the test particle changes when we slightly alter its initial position. By introducing a small displacement, δ_{pos} , to the Lagrange points (L_1 to L_5), we test the stability of these equilibrium points. As noted in Murray & Dermott, the Lagrange points are equilibrium solutions in the synodic frame where the gravitational and centrifugal forces cancel out. However, not all equilibrium points respond the same way to small changes, some are stable, while others are unstable.

The results shown in Figure 10 demonstrate the extreme sensitivity of trajectories to initial conditions in the restricted three-body problem. This phenomenon is a direct consequence of the nature of the Lagrange points as fixed points in the rotating coordinate systems.

According to Murray & Dermott, the collinear points L_1 , L_2 , and L_3 are unstable equilibrium points. They act as saddle points in the potential field. In our simulation, when we add a small displacement (δ_{pos}) to the x -coordinate of these points, the particle does not return to its original position. Instead, the small change grows rapidly as the particle moves away from the equilibrium point. This is clearly visible in the figure, where the trajectories starting near these points shoot off into different regions of the synodic frame. This confirms that these points are mathematically unstable, even the tiniest perturbation prevents the particle from remaining near the point, leading to a trajectory that diverges from the equilibrium.

In contrast, the triangular points L_4 and L_5 exhibit different behavior, provided the mass ratio μ remains below the critical stability threshold (< 0.0385). As discussed in the methodology, these points are centers of stability. When the particle is perturbed near L_4 or L_5 , we expect to see libration or oscillatory motion. Unlike the collinear points, the particle is more likely to remain in the general vicinity of the point, tracing a closed or near-closed orbit rather than flying away immediately. The trajectories in Figure 10 confirm that the motion near the triangular points is significantly more contained than the chaotic divergence observed near the collinear points.

The contrast between these trajectories highlights a fundamental aspect of celestial mechanics: the stability of the three-body system. The sensitivity we observed near the collinear points suggests that in a real-world scenario, a satellite or object placed near L_1 , L_2 , or L_3 would require

active station-keeping maneuvers to prevent it from drifting away. Conversely, the more stable nature of L_4 and L_5 explains why we often observe asteroids (such as the Trojan asteroids in the Sun-Jupiter system) naturally "trapped" in these regions. Our numerical exploration confirms that the choice of initial condition is not just a detail; it determines whether a particle remains bounded in a stable loop or whether it enters a highly unstable, divergent trajectory that may lead to the particle escaping the system entirely.

When we perturb the particle along the x -axis, we are testing the stability against the gravitational gradient. The x -axis is where the gravitational pull of the two primary masses is most dominant. As seen in Figure 10, a small x_0 displacement shifts the particle into a region where the gravitational potential U is significantly different from the equilibrium point. Because L_1, L_2, L_3 are saddle points, an x -perturbation pushes the particle downhill along the potential slope, leading to immediate divergence and escape from the equilibrium. The trajectories effectively roll away from the Lagrange point because the centrifugal force and gravitational attraction no longer balance.

When we perturb the particle along the y -axis, we are testing stability against the Coriolis force. The equations of motion defined by Murray & Dermott include Coriolis terms: $-2n\dot{y}$ and $+2n\dot{x}$. A displacement in y_0 introduces a velocity component that interacts directly with these terms. Unlike the x -perturbation, which primarily tests the gravitational potential slope, a y -perturbation tests the coupling between the particle's motion and the rotating frame. As observed in Figure 10, the trajectories for y -perturbations often exhibit more curved, looping paths compared to the x -perturbations. This is because the Coriolis acceleration turns the particle's path, forcing it into a different orbital trajectory than it would follow in an inertial frame.

The primary difference is that the x -perturbations focus on the energy landscape (the potential U), while the y -perturbations highlight the rotational dynamics (the Coriolis terms).

2) *Initial Velocity*: Because our mass ratio $\mu = 0.01$ is well below the stability threshold ($\mu < 0.0385$) established in Murray & Dermott (Eq. 9), the triangular Lagrange point L_4 is mathematically stable. This means that a particle placed at L_4 will not drift away; instead, it will oscillate around the equilibrium point. These oscillations are known as libration.

Figure 11 illustrate the behavior of 'tadpole orbits', which are the typical trajectories observed near the triangular Lagrange points. When we vary the initial velocity, we are changing the energy level (Jacobi Constant) of the particle, which results in different sizes and shapes of the libration loops.

The $\dot{x}_0$ component represents velocity along the axis connecting the two primary masses. In the synodic frame,

this component is strongly tied to the radial distance from the center of mass. Changing $\dot{x}_0$ shifts the balance between the gravitational pull of the primary bodies and the centrifugal force. As we increase or decrease this velocity, we change the amplitude of the libration. It is easy to noticed that the orbits remain closed loops. This confirms the stability of L_4 . The loops are elongated along the direction of the system's rotation. This is because the Coriolis force ($2n\dot{y}$) acts to turn the particle's motion, preventing it from flying away and forcing it into a stable, enclosed path.

The $\dot{y}_0$ component represents the transverse velocity, which is perpendicular to the x -axis. This velocity component interacts directly with the Coriolis terms in the equations of motion defined in Murray & Dermott (Eqs. 3.16–3.18). Because the Coriolis force is velocity-dependent, changing $\dot{y}_0$ fundamentally alters the phase of the oscillation. By varying $\dot{y}_0$, we are changing the angular momentum of the particle relative to L_4 . This creates a variation in the elliptical shape of the tadpole orbit. It is interesting to see that even with these changes, the orbits stay within a small, defined area.

It is crucial to note that $\dot{x}_0$ and $\dot{y}_0$ affect the system differently due to the anisotropic nature of the synodic frame. Changes $\dot{x}$ are more sensitive to the gravitational gradient. Changes $\dot{y}$ are more sensitive to the Coriolis force. This is why $\dot{y}_0$ variations change the curve or orientation of the loops.

C. Classification of Orbits

To enhance my analysis of the CR3BP, I utilized an interactive simulation tool available [here](#). This enables users to dynamically adjust the mass ratio (μ) and initial state vectors $(x_0, y_0, \dot{x}_0, \dot{y}_0)$ to observe their immediate effects on particle trajectories. By providing real-time visual feedback, this tool bridges the gap between abstract theoretical equations and observable orbital behavior, allowing for the direct exploration of transitions between stable, resonant, and chaotic paths.

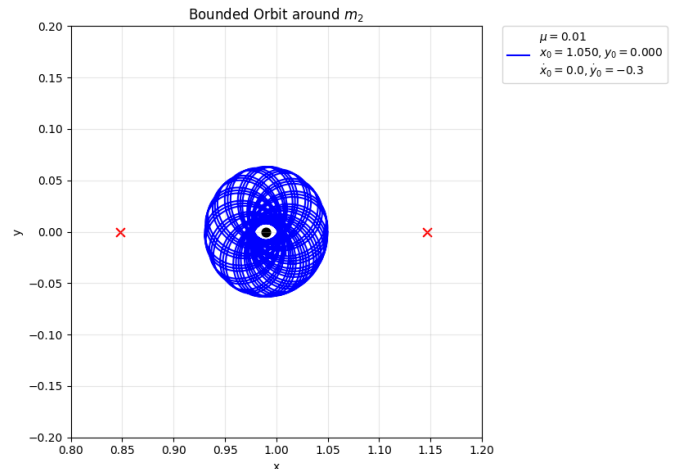


Fig. 2: Bounded Orbit.

Figure 2 shows a bounded orbit around the secondary mass, m_2 . When we set the mass ratio μ to 0.01, the object does not escape into space, instead, it remains trapped by the gravitational pull of m_2 . Looking at the image, we can see that the trajectory creates a dense, circular pattern that looks almost like a flower. This happens because the object orbits the secondary mass while the entire system is rotating. An orbit is bounded when the object stays within a limited distance from the central body. Because the object in this figure completes many loops without moving away, we can say that the initial velocity of $\dot{y}_0 = -0.3$ is perfectly balanced with the gravity of m_2 to keep the satellite contained. The red crosses represent the positions of nearby Lagrange points. The fact that the blue path stays strictly between these points shows that the object is effectively captured by the smaller mass's sphere of influence and does not have enough energy to pass through the L_1 or L_2 gateways to move away.

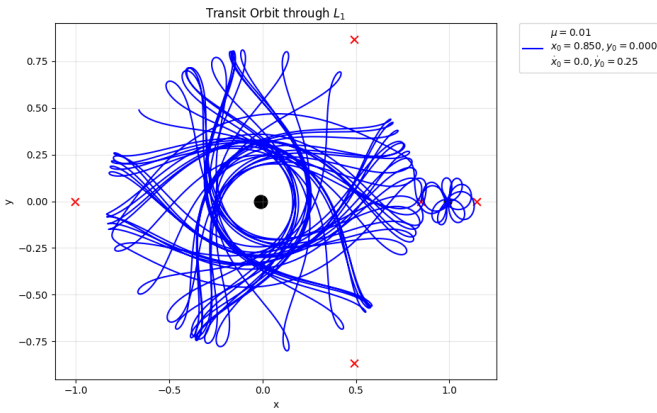


Fig. 3: Transit Orbit.

Figure 3 illustrates a complex transit orbit within a circular restricted three-body problem, defined by a mass ratio of $\mu = 0.01$. Unlike bounded orbits that remain trapped around a single body, this trajectory demonstrates the particle's ability to cross the L_1 Lagrange point, which acts as a gateway between the two primary masses. The path shows the object performing multiple loops around the larger central mass before swinging outwards to orbit the smaller secondary mass, indicating that the initial velocity is sufficient to overcome the gravitational barriers between them. Consequently, the non-repeating, wandering nature of the path highlights a sensitive dependence on initial conditions, even a small change in the starting velocity would significantly alter how the particle transitions between these two regions.

Figure 4 illustrates a horseshoe orbit, which is a unique type of co-orbital motion in a restricted three-body system. In this simulation, using a mass ratio of $\mu = 0.000953$, the particle follows a path that appears to curve around the primary mass (m_1) while being restricted by the orbital position of the secondary mass (m_2). The trajectory forms a distinctive horseshoe shape in the rotating reference frame, effectively bouncing back and forth without ever completing

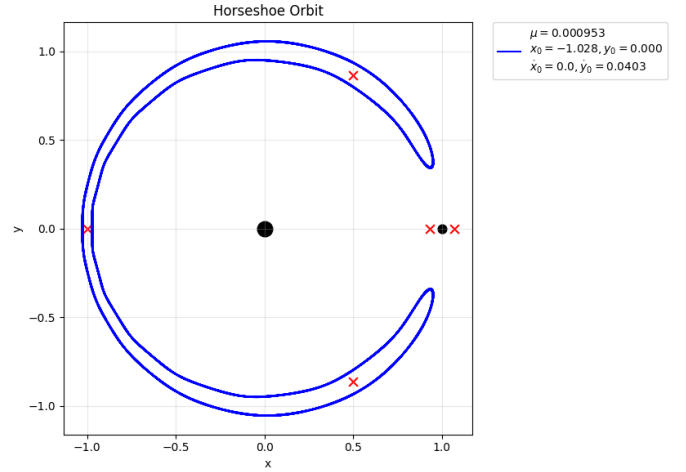


Fig. 4: Horseshoe Orbit.

a full circle around the secondary mass. The object does not have enough energy to pass the gravitational influence of the secondary body, so as it approaches the L_1 and L_2 regions near m_2 , the orbital mechanics cause it to slow down and reverse its direction. This orbit represents a stable long-term relationship where the object stays in a specific configuration relative to the two primary bodies, avoiding collisions while maintaining a predictable, repeating path.

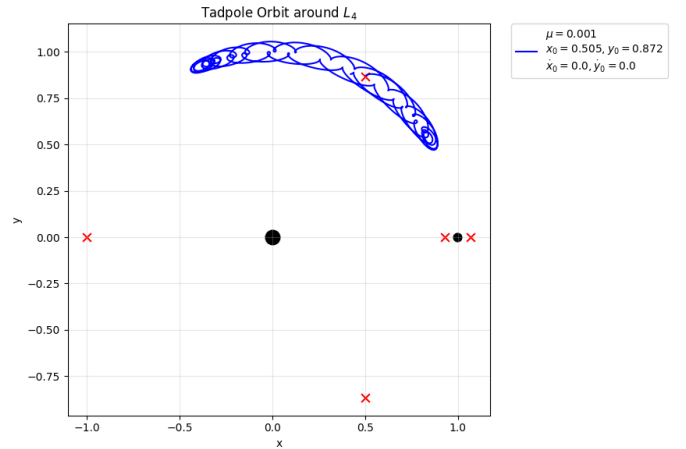


Fig. 5: Tadpole Orbit.

Figure 5 illustrates a tadpole orbit, which is a stable, librating trajectory that oscillates around the L_4 Lagrange point. Unlike transit or escape trajectories that travel through the entire system, this path is strictly confined to the region near the L_4 point, where the gravitational forces from the two primary bodies effectively balance the motion of the particle. As shown by the blue trajectory, the object performs a series of small, repeating loops that create a distinct, kidney-shaped pattern while it moves back and forth around the equilibrium point. This behavior, known as libration, occurs because the particle is caught in a stable gravitational zone, consequently, its relative position remains consistent over long periods of time. Because the

object neither drifts away into space nor crashes into the primary masses, the tadpole orbit serves as a perfect example of long-term stability within the restricted three-body problem.

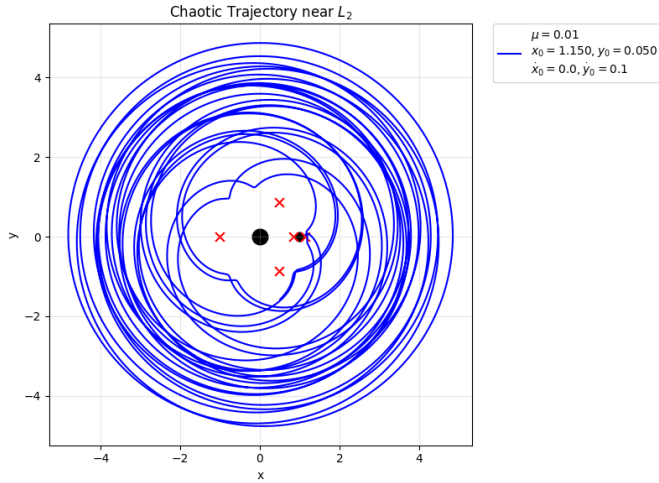


Fig. 6: Chaotic Trajectory.

Figure 6 displays a chaotic trajectory. To understand this motion, it is helpful to contrast it with the stable orbits, like the tadpole or horseshoe orbits, that we discussed previously. This trajectory is a classic example of deterministic chaos. With a mass ratio (μ) of 0.01, the particle does not settle into a periodic, closed loop. Instead, it exhibits a sensitive dependence on its starting position and velocity, meaning that even a microscopic change in the initial parameters would result in a completely different path. The trajectory is centered near the L_2 Lagrange point. Because L_2 is an unstable equilibrium point, particles in this region are strongly influenced by what are known as unstable manifolds. Rather than being caught in a stable resonant zone, the particle is pushed and pulled between different gravitational regimes, causing it to wander through the phase space. Unlike the horseshoe or tadpole orbits, which are locked into resonance with the primary masses, this chaotic path has no commensurability between its orbital frequency and the rotation of the system. Consequently, the particle traces out a dense, irregular loops, effectively filling the available space rather than following a predictable, repetitive track.

Figure 7 illustrates an escape trajectory, where a particle has enough kinetic energy to leave the gravitational influence of the two primary masses entirely. Unlike the bounded or resonant orbits we examined earlier, this path shows the particle spiraling outward, never to return to the center. The key difference in this simulation is the initial velocity, $\dot{y}_0 = 1.5$. By providing this higher energy, the particle gains enough speed to break free from the gravitational well of the system. In the CR3BP, we use the Jacobi Constant (C_J) to define allowed regions. At this energy level, the Zero Velocity Curves, the barriers that usually trap particles in bounded or tadpole orbit,—are effectively open, allowing the particle to move to infinity. In the rotating reference frame, the path

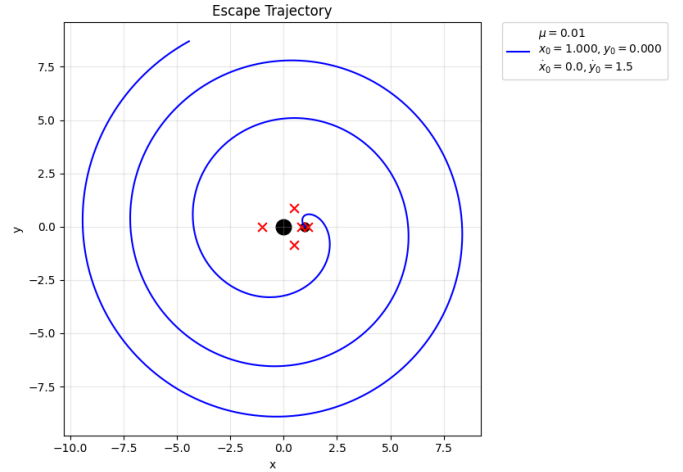


Fig. 7: Escape trajectory.

appears as an expanding spiral. This happens because as the particle moves further away, the centrifugal and Coriolis forces of the rotating system influence its path, causing it to curve outward rather than flying in a perfectly straight line. Because the particle is moving away from the system, it will eventually enter a region where the gravitational pull of the two primary bodies becomes negligible. At this point, the particle's motion will become dominated by its own inertia, moving away at a constant velocity in an inertial frame.

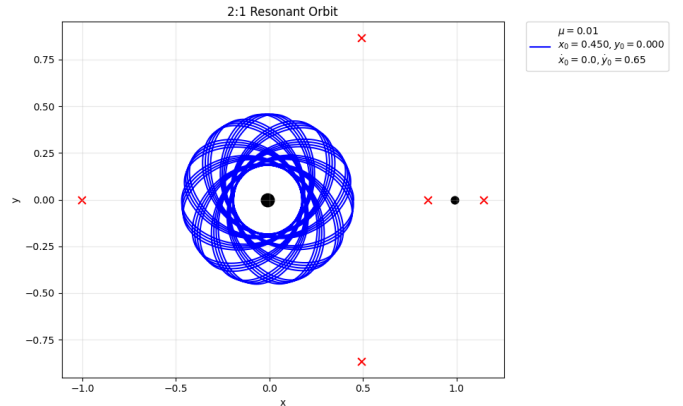


Fig. 8: Resonant Orbit.

Figure 8 shows a resonant orbit, a special type of trajectory in the CR3BP where the object's orbital motion is synchronized with the rotation of the system. When the orbital periods of the particle and the primary masses are related by a ratio of small integers, the gravitational interactions repeat in a predictable, stable, and periodic pattern. Unlike the chaotic trajectories we have discussed, this path is closed. This means that after a certain number of revolutions, the particle returns to its exact starting point in the rotating reference frame. The 2:1 ratio indicates that for every single orbit of the primary bodies in the system, the particle completes exactly two full loops. This perfect synchronization is why the pattern looks like a symmetrical, repeating geometric flower. Gravitational balance

occurs because the particle is placed in a specific region where the combined gravitational pull of the two primaries and the centrifugal forces of the rotating frame perfectly cancel out at regular intervals, preventing the object from drifting away or colliding with the primaries.

D. Stability Analysis

To determine the stability of the Lagrange points, we conducted a numerical experiment observing how the system responds to a small displacement (or "nudge") from equilibrium. By systematically varying the mass ratio μ across values of 0.025, 0.035, and 0.045, we can test the theoretical stability criterion:

$$\mu < \frac{1 - \sqrt{23/27}}{2} \approx 0.0385$$

This inequality defines the threshold below which the triangular points (L_4 and L_5) are considered stable. We simulated trajectories starting at $(L_x + 10^{-4}, L_y + 10^{-4})$ to observe if the particle remains in the vicinity of the Lagrange point or diverges from it.

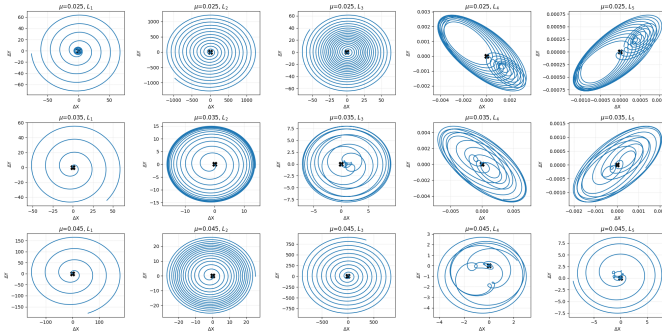


Fig. 9: Evolution of trajectories near the five Lagrange points (L_1 to L_5) following a small perturbation, compared across mass ratios $\mu = 0.025, 0.035$, and 0.045 . The black X marks the original equilibrium position.

The results presented in Figure 9 allow us to distinguish between two fundamentally different types of dynamic behavior: the inherent instability of the collinear points and the conditional stability of the triangular points.

For collinear points L_1, L_2, L_3 , as predicted by the theoretical foundations in Murray & Dermott, the collinear points exhibit instability regardless of the mass ratio μ . In all plots for L_1, L_2 , and L_3 , we observe that the trajectory quickly moves away from the origin ($\Delta X = 0, \Delta Y = 0$). These points act as saddle points in the gravitational potential landscape. Once a particle is nudged, even slightly, the gravitational forces from the two primary masses do not act to return it to the point. Instead, the particle slides away from the equilibrium, which is reflected in our plots as a rapid increase in ΔX and ΔY over time. This instability is a consistent feature and does not depend on the specific value of μ .

The behavior of the triangular points L_4, L_5 depends strictly on the value of the mass ratio μ . For $\mu < 0.0385$, our simulations show that the particle remains trapped in a small, closed loop around the Lagrange point. This is known as libration. Even after a perturbation, the Coriolis force interacts with the particle's velocity to produce a restoring effect, keeping the particle near the triangular point. This confirms that for these mass ratios, the triangular points are stable equilibrium locations. When $\mu > 0.0385$, the plots show a clear departure from the closed-loop behavior seen in the stable cases. Instead of orbiting the point, the trajectory begins to spiral away from the origin. This represents the transition where the triangular point loses its stability. Mathematically, this is the point where the eigenvalues of the linearized equations of motion change from purely imaginary (stable oscillations) to having real parts (exponential divergence).

IV. CONCLUSION

In conclusion, this project provided a detailed analysis of the Circular Restricted Three-Body Problem. By integrating the equations of motion, we have gained a clearer understanding of how a particle moves within a system containing two large gravitational bodies. The results demonstrate that the Jacobi constant is the most critical factor in determining the behavior of a particle; as energy levels change, the available paths for the particle shift from free-roaming motion to restricted, trapped orbits. Furthermore, our classification of these paths, ranging from resonant and tadpole orbits to chaotic and escape trajectories, highlights the complex relationship between initial conditions and long-term stability. The stability analysis of the triangular Lagrange points further confirms that these areas are essential for maintaining stable positions within space systems. Ultimately, these findings are highly valuable for space mission design, as they allow us to predict which orbits are reliable and how we can use natural gravitational gateways to move spacecraft efficiently. This study reinforces the fact that orbital mechanics is not random, but follows precise physical rules that allow for the careful planning of future space exploration.

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- [1] Murray, C. D., & Dermott, S. F. (1999). Solar system dynamics. Cambridge University Press.

APPENDIX

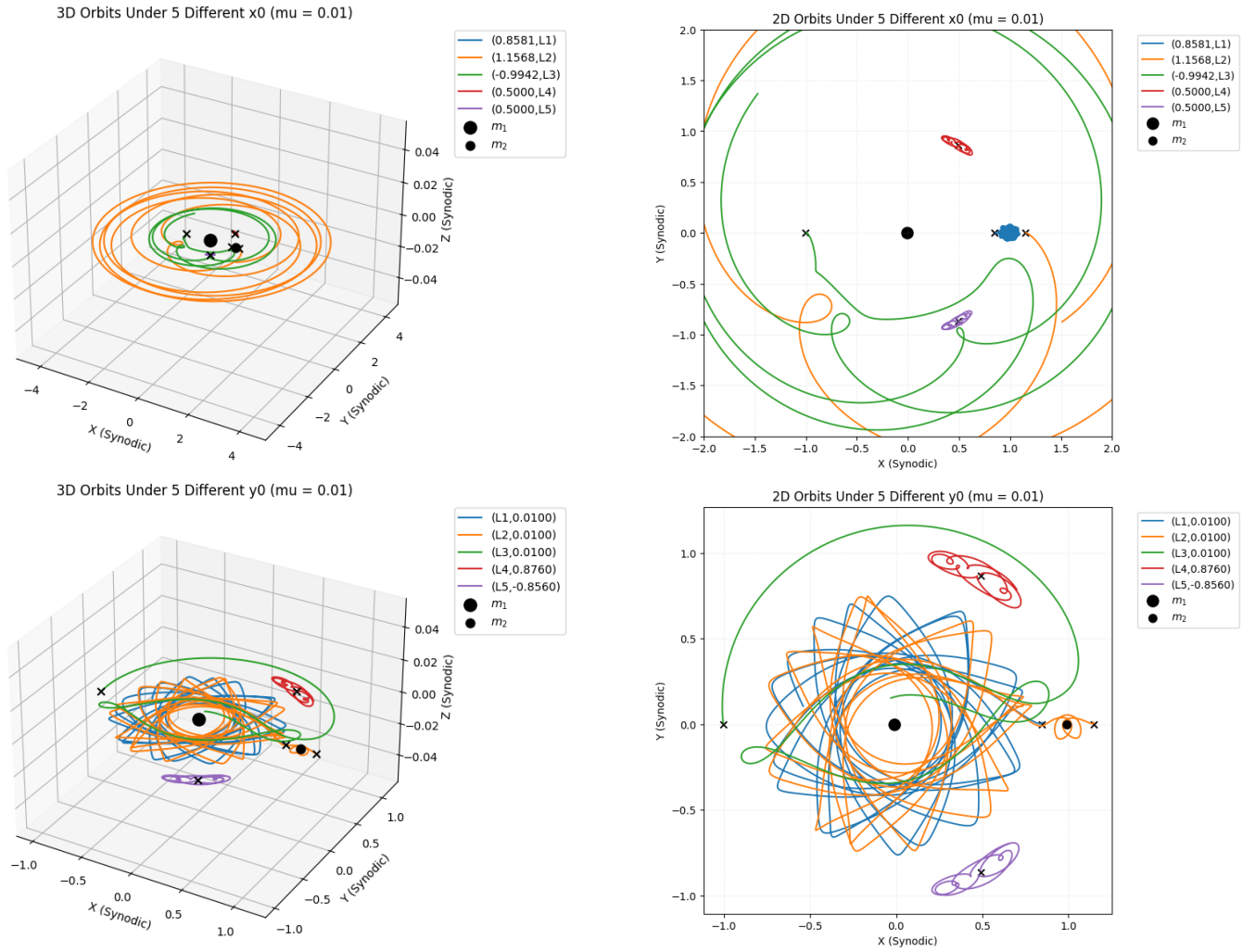


Fig. 10: 3D and 2D visualization of trajectory sensitivity for initial conditions perturbed by δ_{pos} near the five Lagrange points. The black crosses represent the original equilibrium Lagrange points, while the colored lines show the diverging paths of the particle after a small displacement in the positions.

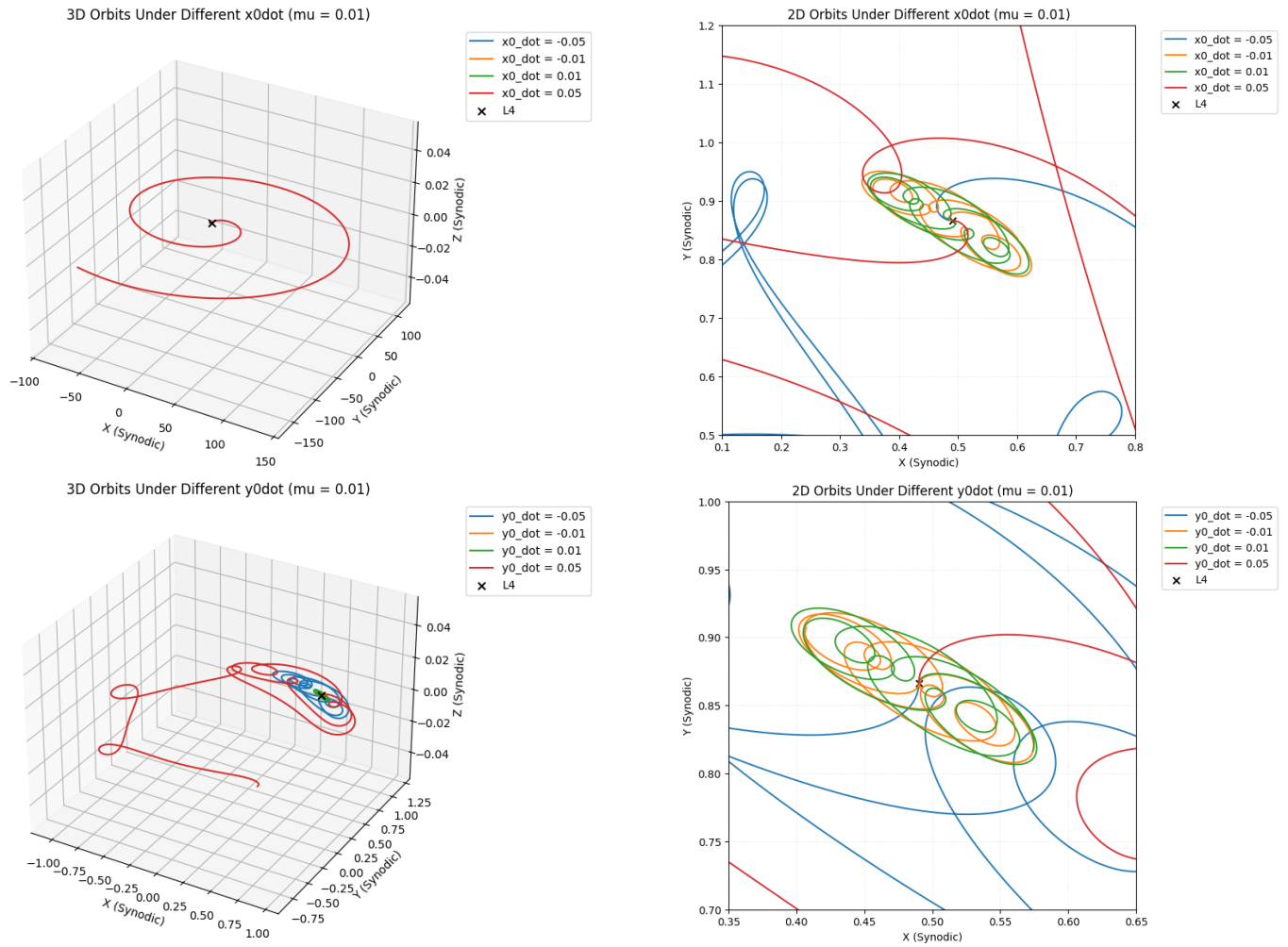


Fig. 11: 3D and 2D visualization of trajectory sensitivity for initial conditions perturbed by δ_{vel} near L_4 . Each line represents a slightly different initial energy state. Note how the different transverse momentum changes the shape and orientation of the oscillation loops.